

Representation Separability of Signal-to-Image Transformations for Sleep Stage Classification Using Pretrained Vision Transformers

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Abstract: Accurate sleep stage classification is essential for diagnosing sleep disorders, guiding treatment decisions, and advancing the understanding of brain function during sleep. Manual scoring of polysomnography (PSG) remains the clinical gold standard but is time-consuming and prone to inter-rater variability, motivating the need for reliable automated methods. While convolutional neural networks have demonstrated strong performance in this domain, Vision Transformers (ViTs) remain underexplored despite their ability to capture global dependencies. The primary objective of this study is to systematically evaluate the effectiveness of different signal-to-image (S2I) transformations for ViT based sleep staging and to identify strategies that yield the most discriminative representations. This study systematically evaluated 1,016 combinations of S2I methods, fusion strategies, and channel selections using the CAP Sleep Database, comprising 108 overnight PSG recordings annotated into standard 30-second sleep epochs. After preprocessing and artifact removal, four S2I transformations spectrograms, GASF, GADF, and MTF were applied, with fusion achieved through grid tiling or channel stacking. Embeddings extracted from a finetuned ViT-B/16 served as the basis for assessing stage separability using Fisher's Discriminant Ratio, Silhouette Coefficient, Davies-Bouldin Index, and Calinski Harabasz Index. Composite scores derived from these metrics ranked strategies, and the top were validated through classification benchmarks with several base ViT models. Results showed that GASF with grid tiling across all channels produced the most discriminative representations (composite score = 3.95), significantly outperforming other strategies (Mann Whitney U, $p < 0.001$, $r = 0.68$, where p denotes significance level and r denotes rank-biserial correlation effect size). This top-ranked transformation was further assessed using five Vision Transformer architectures ViT B/32, DeiT-B, Swin-B, BEiT-B, and PVT-Small under five-fold stratified cross-validation. Performance was evaluated by overall accuracy and macro-averaged F1-score. ViT-B/32 achieved the highest results (83.1% accuracy, 79.4% macro-F1), with Swin-B and BEiT-B performing closely. These findings confirm that effective S2I strategies enhance ViT-based sleep staging and underscore the clinical potential of Vision Transformers.

Keywords: Sleep Stage Classification, Vision Transformers, Signal-to-Image Transformation, Deep Learning